



# Strength performance and environmental assessment of Portland clinker

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## ABSTRACT

Alite influences both the early-age strength development and environmental footprint of concrete. However, the link between the alite content in clinker, strength development in concrete, and its carbon footprint remains overlooked in life cycle assessment. We study CEM I mortars containing clinkers with 5–85 wt% alite, combining thermodynamic modelling with a compressive strength prediction model based on the combined water fraction. Our model predicts that mortars made from high alite clinkers have 29% higher 28-day compressive strengths than that of low alite clinkers, at equivalent clinker and water contents. However, higher alite clinkers increase the CO<sub>2</sub>-eq. emissions of mortars by up to 19% compared with lower alite clinkers. In general, increasing alite content in clinker reduces CO<sub>2</sub>-eq. emissions per unit of strength gained in mortar. Therefore, the carbon footprint of concrete can be reduced by optimising both the alite content in clinker and concrete mix design.

## 1. Introduction

The cement industry currently contributes around 8% of global CO<sub>2</sub> emissions [1], predominantly due to Portland cement clinker (hereafter clinker) production. Clinker is produced as nodules in a rotary kiln, where limestone is calcined ( $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ ) with siliceous and argillaceous materials such as clay at  $\sim 1450^\circ\text{C}$ . The nodules are then rapidly cooled to preserve the high temperature phases and then ground into a powder. Clinker is a heterogeneous solid material formed of impure calcium silicate phases such as alite ( $\text{C}_3\text{S}^1$ ) and belite ( $\text{C}_2\text{S}$ ). These phases dissolve in water, and precipitate lower-density, strength-giving hydrate phases, notably calcium silicate hydrate (C-S-H). Within clinker phases, alite is the more reactive one, responsible for most of the early-age strength gain in concrete [2], and contains the most Ca by mass of the clinker phases. The typical clinker oxide composition is  $\sim 67$  wt% CaO, 22 wt% SiO<sub>2</sub>, 5 wt% Al<sub>2</sub>O<sub>3</sub>, and 3 wt% Fe<sub>2</sub>O<sub>3</sub> (balance 3 wt% other components) [3]. The clinker phase composition is more variable (Table 1) and depends on factors such as kiln conditions and nodule cooling rate.

Approximately 60% of the CO<sub>2</sub>-eq. emissions from clinker

production arise from the calcination of limestone, which is the typical and widely available Ca source for this process. Fossil fuel combustion to supply process heat typically contributes  $\sim 35\%$  of clinker CO<sub>2</sub>-eq. emissions. The remaining  $\sim 5\%$  CO<sub>2</sub>-eq. emissions are mainly from electrical processes (e.g., grinding kiln feedstocks and clinker) [7]. Due to the high emissions associated with clinker production, clinker substitution is one of the most effective means of cement decarbonisation [8,9], and much research attention (e.g., [9–12]) focusses on producing composite cements with lower-carbon footprint materials such as blast furnace slag, calcined clay, and limestone. However, clinker substitution extents of  $>50$  wt% in cement are generally limited by the reduced reactivity of the resulting composite cements (from reduced alite content) which negatively affects the cement strength class. Clinker reactivity has been identified as the main technical limitation to improve composite cement performance and hence further decarbonisation [2].

The clinker phase composition is therefore an important variable for both the compressive strength development (i.e., ‘cement strength performance’) and carbon footprint of concrete. However, clinker composition has been largely unexplored in life cycle assessment (LCA) studies of cement and concrete [13]. Many LCA models for Portland cement

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<sup>1</sup> Using cement chemistry notation, where C = CaO, S = SiO<sub>2</sub>, A = Al<sub>2</sub>O<sub>3</sub>, F = Fe<sub>2</sub>O<sub>3</sub>. C<sub>3</sub>S refers to the pure form of the alite phase (tricalcium silicate) and C<sub>2</sub>S refers to the belite phase (dicalcium silicate). Other main clinker phases discussed in this paper include aluminat (tricalcium aluminate, C<sub>3</sub>A) and ferrite (C<sub>4</sub>AF). Here, we use the terms alite, belite, aluminat, and ferrite as they more accurately reflect their nonzero impurity contents in clinker.

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**Table 1**

This table shows the typical phase composition range of clinker from sources within the literature.

| Source | Alite (wt%) | Belite (wt%) | Aluminate (wt%) | Ferrite (wt%) |
|--------|-------------|--------------|-----------------|---------------|
| [4]    | 56–74       | 3–28         | 2–14            | 2–17          |
| [5]    | 37–81       | 4–35         | 1–16            | 0–16          |
| [6]    | 56–74       | 4–25         | 1–8             | 7–14          |

(PC) exist, typically focussing on different kiln arrangements and fuel sources, but use a unit mass of cement as the functional unit (e.g., kg CO<sub>2</sub>-eq. per kg cement) [7,14–16]. Mass-based functional units are also used in LCA studies of cement that consider clinker chemistry. For example, Boesch et al. used a mass-based functional unit for their LCA of clinker production, but limited process changes (e.g., raw meal and fuel changes) to avoid affecting clinker composition [17]. Miller and Myers used a mass-based functional unit, evaluating the environmental performance of alternative clinker chemistries and composite cements largely based on the heats of reaction in each system [18]. Cao et al. developed a life-cycle model linking clinker composition and carbon footprint from two cement production lines but did not account for cement strength performance. They concluded that “lower quality” (lower alite) clinkers should be targeted to reduce clinker emissions [19], in direct conflict with the assertion by Scrivener et al. that high belite clinkers (lower alite) do not offer sufficient performance to justify their lower emissions [2].

The use of performance-based functional units in LCA of concrete is also relatively uncommon. Damini et al. surveyed the literature to quantify the ‘binder intensity’, the amount of binder in a concrete mix per 1 MPa of 28-days compressive strength, and the ‘CO<sub>2</sub> intensity’, the CO<sub>2</sub>-eq. emissions footprint of a concrete mix per 1 MPa of 28-days compressive strength [20]. Abrao et al. experimentally evaluated the environmental impact of different composite cements using a strength-based functional unit [21,22]. They found that cement strength performance in mortars was strongly correlated with the combined water fraction, which is the amount of chemically bound water in hydration products (i.e., ‘combined water’) divided by the mixing water in mortar. Therefore, the combined water fraction enables prediction of cement strength performance, which reflects the degree of hydration and cement mix design (e.g., w/c ratio). However, experimentally determining the combined water fraction across the large design space of concretes is a significant burden because many cement and concrete types exist.

Heeren and Myers [23] proposed a computational framework to explore the possible design space of composite cements, developing the model ‘Panoramix’ to integrate cement property prediction (e.g. cement strength performance) and LCA. Their work was recently updated with a focus on durability assessment [24]. Panoramix uses an established thermodynamic modelling approach for cementitious systems (e.g., [25]) to predict the hydrated cement phase assemblage under different mix designs and curing conditions. However, for simplicity, these initial

versions of Panoramix treat clinker as a congruently dissolving mineral, and so do not capture the effects of clinker composition (e.g., higher reactivity of alite) [24]. Advancing this approach by coupling the combined water fraction with solid phase assemblage prediction from thermodynamic modelling could be a powerful tool for strength performance-based environmental assessment of different clinkers, cements, mortars, and concretes.

This study investigates the effect of clinker alite content on cement strength performance and carbon footprint. We focus on the calcination and fuel related CO<sub>2</sub>-eq. emissions of clinker production. Cement strength performance is evaluated by modifying the thermodynamic approach proposed by Heeren and Myers [23] and updated by Gao et al. [24] to include incongruent clinker phase dissolution, coupling the output with the empirical cement strength performance correlation based on combined water fraction proposed by Abrao et al. [21,22].

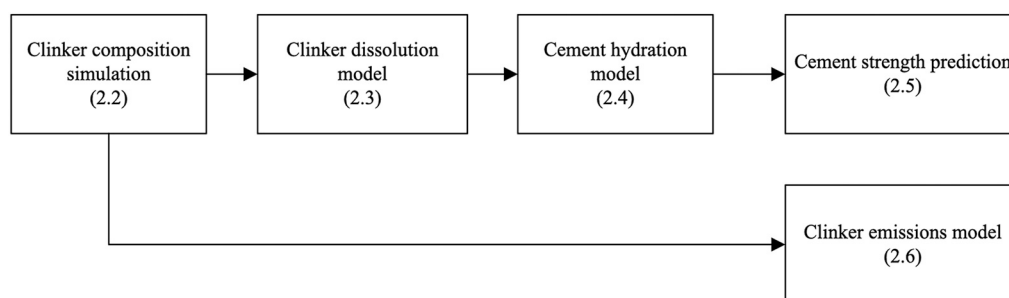
## 2. Method

### 2.1. Model overview

The scope of this work is to produce a cement strength performance-based environmental assessment of CEM I cement mixes (i.e., Portland cements with >95 wt% clinker per BS EN 197-1 [26]) made with clinkers of varying alite content. We model the compressive strength at 28-days curing for 1:3 (cement:sand) mortars with an 0.5 w/c ratio per typical strength tests (e.g., [27]). Fig. 1 below provides an overview of the multi-step modelling approach used in this work.

### 2.2. Clinker composition simulation

A synthetic dataset (Supporting Information S2 Table 1 (clinkers) Table 2 (cements)) was created using Python (version 3.11) with 500 simulated clinker compositions, where clinker alite content varied between 5 and 85 wt%. In an initial step, a random distribution package (random.uniform) was used to generate clinker compositions between 5–95 g alite; 5–95 g belite; 4–5 g aluminate; and 4–5 g ferrite, per 100 g clinker. We favoured brute force random sampling to obtain a large sample set without a biased distribution of clinker phases, allowing strength trends to emerge from chemically feasible compositions representative of industrial production to improve the generalizability of the results. The sum of alite, belite, aluminate, and ferrite was constrained between 95 and 100 g by excluding simulated clinkers outside this range. Up to 5 g of an ‘inert’ phase, which does not participate in cement hydration, was then added such that there was 100 g of each simulated clinker, to avoid distorting the randomly sampled uniform distribution. CEM I cements were generated for each clinker using a mix design of 100 g clinker to 5 g gypsum. No additional minor constituents (e.g., limestone) were added to the CEM I cements to maximise the amount of clinker in cement, discussed further in Supporting Information S1, S5.0. The 105 g (clinker + gypsum) of material was then



**Fig. 1.** Schematic overview of modelling approach. Boxes represent the stages in the model, with interdependencies tracked by arrows. Dotted arrow feeding into strength prediction to infer only w/c ratio is required for the cement strength prediction in addition to the hydration model results. Numbers in brackets e.g., (2.2) refers to the methodology subsection where the modelling stage is described in detail.

**Table 2**

Parrot and Killoh constants for clinker phases from [29,30].

|           | $K_1$ | $N_1$ | $K_2$ | $K_3$ | $N_3$ | $H$  | $E_a$ (kJ mol <sup>-1</sup> ) |
|-----------|-------|-------|-------|-------|-------|------|-------------------------------|
| Alite     | 1.5   | 0.7   | 0.05  | 1.1   | 3.3   | 2    | 41.6                          |
| Belite    | 0.5   | 1     | 0.02  | 0.7   | 5     | 1.55 | 20.8                          |
| Ferrite   | 1     | 0.85  | 0.04  | 1     | 3.2   | 1.8  | 54.0                          |
| Aluminate | 0.37  | 0.7   | 0.015 | 0.4   | 3.7   | 1.65 | 34.1                          |

normalised to 100 g of cement.

### 2.3. Clinker dissolution model

Incongruent dissolution of clinker in CEM I cement paste at 28-days of curing was modelled using the Parrot and Killoh dissolution rate model [28]. The dissolution rate for each phase (alite, belite, aluminate, and ferrite) at time  $t$  and temperature  $T$  ( $R_{t,T}$ ) was defined by Eq. (1). The hydration degree ( $\alpha_t$ ) was defined per Eq. (2). We aligned the model parameters with the EN 196-1 testing standard for determining compressive strength in mortar samples [27], which specifies a w/c mass ratio of 0.5 and temperature of 293.15 K. Other global parameters include: surface area, 413 m<sup>2</sup> kg<sup>-1</sup>; reference temperature ( $T_0$ ), 293.15 K; and relative humidity (RH), 100%. The reacted amount of each clinker phase at 28-days curing was calculated by the hydration degree multiplied by the phase mass fraction in each CEM I cement (g phase per 100 g cement). We assumed 100% of gypsum was dissolved at 28-days curing [29], and that the contribution of sand in mortar on hydration of cement phases was negligible.

$$R_{t,T} = \min \left\{ \frac{K_1}{N_1} (1 - \alpha_t) (-\ln(1 - \alpha_t))^{(1-N_1)} \cdot \left( \frac{\text{surface area}}{385} \right), \frac{K_2(1 - \alpha_t)^{\frac{2}{3}}}{1 - (1 - \alpha_t)^{\frac{1}{3}}}, K_3(1 - \alpha_t)^{N_3} \right\} \cdot \left( \frac{rh - 0.55}{0.45} \right)^4 \cdot e^{-\frac{E_a}{R} \left( \frac{1}{T} - \frac{1}{T_0} \right)} \quad (1)$$

$$\alpha_t = \begin{cases} \alpha_{t-1} + \Delta t \cdot R_{t-1}, \\ \alpha_{t-1} + \Delta t \cdot R_{t-1} \cdot \left( 1 + 3.33 \left( H \cdot \frac{w}{c} - \alpha_t \right) \right)^4, \end{cases} \quad \begin{cases} \text{if initial rate } \alpha_t < H \cdot \frac{w}{c} \\ \text{if } \alpha_t \geq H \cdot \frac{w}{c} \end{cases} \quad (2)$$

Constants for each phase (e.g.,  $K_1$ ,  $K_2$ ,  $K_3$ ,  $N_1$ ,  $N_3$ ) and phase activation energies ( $E_a$ ) were taken from Lothenbach et al. [29,30] based on experimental observations for OPC clinker (Table 2).

The degree of hydration of each phase at 28-days curing is summarised in Supporting Information S1, S1.0. As the constants for each phase are derived experimentally from OPC clinkers, we validated the applicability of the predicted degree of hydration for high-alite and high-belite clinkers. The Parrot and Killoh constants are well adapted for high-alite clinkers, and represents ‘best-case’ dissolution performance for high-belite clinkers compared to reported literature belite dissolution. Therefore, the use of this model is considered appropriate and conservative for the current analysis. Further discussion is available in Supporting Information S1, S1.0.

### 2.4. Thermodynamic hydration model

We predicted the binder chemistry at 28-days curing for 100 g CEM I cement, 50 g water, and 0.01 g of O<sub>2</sub> using GEMS Selektor v.3 software [31]. Unreacted and inert clinker phases were not included. To improve the generalisability of the model results and estimate the envelope of hydration products, two models for the C-S-H phase were tested: CSHQ [32] and CNASH<sub>ss</sub> [33]. The CSHQ model is well adapted for PC systems where the C-S-H phase has a high Ca/Si molar ratio [25], but does

not predict aluminium uptake in C-S-H (e.g., as C-(A-)S-H). Aluminium uptake is particularly important for composite cements [25]. The CNASH<sub>ss</sub> model incorporates aluminium uptake in C-S-H, but underestimates C-(A-)S-H content (and overestimates portlandite content) at Ca/Si molar ratios >1.3 [25]. Considering both CSHQ and CNASH<sub>ss</sub> models captures a wider range of potential end-members for C-S-H and thus define an ‘envelope’ of results encompassing potential variations in the C-S-H composition and hydrate water content. We refer to all C-S-H phases, including C-(A-)S-H as predicted by CNASH<sub>ss</sub>, as C-S-H for simplicity. All other cement-specific data used were per the CEM-DATA18 database [33,25].

From the CEMDATA18 data, we selected the tricarboalu03 and ettringite03<sub>ss</sub> models for AFt and the C4AH13 and monosulphate12 models for AFm; the solid solution models ettringite-AlFe, ettringite-FeAl, monosulph-AlFe, and monosulph-FeAl were disabled. Phases kinetically unlikely to form at 28-days of curing (gibbsite, kaolinite, hematite, goethite, and quartz) were disabled. No dissolved species were disabled in the aqueous solution.

### 2.5. Combined water fraction and strength prediction

The amount of combined water at 28-days curing for each CEM I paste was determined stoichiometrically from the amount of water in each precipitated hydrate phase (Table 3).

C-S-H has a diverse composition depending on the aqueous solution chemistry [34,35]. Reported C-S-H combined water contents from the literature are shown in Table 4.

The combined water fraction (cwf) was calculated per Eq. (3) using the combined water ( $w_c$ , based on the molar amount of each hydrate

phase ( $n_i$ , mol) and stoichiometry ( $n_{H_2O,i}$ , mol), and molar mass of water ( $M_w(\text{H}_2\text{O}) = 18 \text{ g mol}^{-1}$ ), and the mixing water ( $w_m$ ) of 50 g.

$$cwf = \frac{w_c}{w_m} = \frac{18 \cdot \sum_i n_i \cdot n_{H_2O,i}}{50} \quad (3)$$

We validated the calculated combined water results by comparing to the experimental combined water results reported in [22] for a CEM I paste (66.1 wt% alite content in clinker, 3.5 wt% gypsum in cement, 4.9 wt% limestone in cement), where hydration was first arrested using solvent exchange [39] and then lightly dried at 50 °C before the combined water was determined from the mass loss between 50 °C and 550 °C (e.g., per protocol [40,41]). Note that Abrao et al. use cement paste to characterise the combined water, and correlate this to the mixing water and compressive strength of 1:3 (cement:sand) prismatic mortar specimens. This is because the combined water is not significantly affected by the mixing water, provided sufficient water is available for hydration [42]. We also calculated the combined water for the same mix using the thermodynamic model (Section 2.4) and CEM-DATA18 stoichiometry (Table 3). Further details of the validation process are given in Supporting Information S1, S4.0. The cement strength performance ( $\sigma$ , MPa) at 28-days curing was then determined using the empirical correlation derived from compressive strength test results of mortar samples with a 1:3 cement to sand ratio as given in [22] (Eq. (4)).

**Table 3**

Stoichiometry and water content (in mol H<sub>2</sub>O per mol phase) of hydrate phases per by cemdata18. Endmember names and molecular formulae are given corresponding to the models in the CEMDATA18 database [25]. C-S-H end members are grouped by model; both the CSHQ and CNASH\_SS models in CEMDATA18 were used for C-S-H prediction.

| Phase            | End member<br>(per cemdata18)                       | Molecular formula<br>(per cemdata18)                                                                                                                                                      | mol H <sub>2</sub> O |
|------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| AFm              | Monosulphate12                                      | Ca <sub>4</sub> Al <sub>2</sub> SO <sub>10</sub> (H <sub>2</sub> O) <sub>12</sub>                                                                                                         | 12                   |
|                  | C <sub>4</sub> AH <sub>13</sub>                     | Ca <sub>4</sub> Al <sub>2</sub> (OH) <sub>14</sub> (H <sub>2</sub> O) <sub>6</sub>                                                                                                        | 6                    |
| Hydrotalcite     |                                                     | Mg <sub>4</sub> Al <sub>2</sub> O <sub>7</sub> (H <sub>2</sub> O) <sub>10</sub>                                                                                                           | 10                   |
| AFt              | tricarboalu03                                       | (CO <sub>3</sub> )Ca <sub>2</sub> Al <sub>0.67</sub> (OH) <sub>4</sub> (H <sub>2</sub> O) <sub>8.67</sub>                                                                                 | 8.67                 |
|                  | ettringite03_ss                                     | (SO <sub>4</sub> )Ca <sub>2</sub> Al <sub>0.67</sub> (OH) <sub>4</sub> (H <sub>2</sub> O) <sub>8.67</sub>                                                                                 | 8.67                 |
| Hydro-garnet     | C <sub>3</sub> AFS <sub>0.8</sub> H <sub>4.32</sub> | (AlFe <sub>1/3</sub> O <sub>3</sub> )[Ca <sub>3</sub> O <sub>3</sub> (SiO <sub>2</sub> ) <sub>0.84</sub> (H <sub>2</sub> O) <sub>4.32</sub> ]                                             | 4.32                 |
|                  | C <sub>3</sub> FS <sub>0.8</sub> H <sub>4.32</sub>  | (Fe <sub>1/3</sub> Fe <sub>1/3</sub> O <sub>3</sub> )[Ca <sub>3</sub> O <sub>3</sub> (SiO <sub>2</sub> ) <sub>0.84</sub> (H <sub>2</sub> O) <sub>4.32</sub> ]                             | 4.32                 |
| Portlandite      |                                                     | Ca(OH) <sub>2</sub>                                                                                                                                                                       | 1                    |
| C-S-H (CNASH_ss) | 5CA                                                 | (CaO) <sub>1.25</sub> (SiO <sub>2</sub> ) <sub>1</sub> (Al <sub>2</sub> O <sub>3</sub> ) <sub>0.125</sub> (H <sub>2</sub> O) <sub>1.625</sub>                                             | 1.63                 |
|                  | INFCA                                               | (CaO) <sub>1</sub> (SiO <sub>2</sub> ) <sub>1.1875</sub> (Al <sub>2</sub> O <sub>3</sub> ) <sub>0.15625</sub> (H <sub>2</sub> O) <sub>1.65625</sub>                                       | 1.66                 |
|                  | 5CNA                                                | (CaO) <sub>1.25</sub> (SiO <sub>2</sub> ) <sub>1</sub> (Al <sub>2</sub> O <sub>3</sub> ) <sub>0.125</sub> (Na <sub>2</sub> O) <sub>0.25</sub> (H <sub>2</sub> O) <sub>1.375</sub>         | 1.38                 |
|                  | INFCNA                                              | (CaO) <sub>1</sub> (SiO <sub>2</sub> ) <sub>1.1875</sub> (Al <sub>2</sub> O <sub>3</sub> ) <sub>0.15625</sub> (Na <sub>2</sub> O) <sub>0.34375</sub> (H <sub>2</sub> O) <sub>1.3125</sub> | 1.31                 |
|                  | INFCN                                               | (CaO) <sub>1</sub> (SiO <sub>2</sub> ) <sub>1.5</sub> (Na <sub>2</sub> O) <sub>0.3125</sub> (H <sub>2</sub> O) <sub>1.1875</sub>                                                          | 1.19                 |
|                  | T2C-CNASHss                                         | (CaO) <sub>1.5</sub> (SiO <sub>2</sub> ) <sub>1</sub> (H <sub>2</sub> O) <sub>2.5</sub>                                                                                                   | 2.5                  |
|                  | T5C-CNASHss                                         | (CaO) <sub>1.25</sub> (SiO <sub>2</sub> ) <sub>1.25</sub> (H <sub>2</sub> O) <sub>2.5</sub>                                                                                               | 2.5                  |
|                  | TobH-CNASHss                                        | (CaO) <sub>1</sub> (SiO <sub>2</sub> ) <sub>1.5</sub> (H <sub>2</sub> O) <sub>2.5</sub>                                                                                                   | 2.5                  |
|                  | CSHQ-JenD                                           | (CaO) <sub>1.5</sub> (SiO <sub>2</sub> ) <sub>0.6667</sub> (H <sub>2</sub> O) <sub>2.5</sub>                                                                                              | 2.5                  |
|                  | CSHQ-JenH                                           | (CaO) <sub>1.3333</sub> (SiO <sub>2</sub> ) <sub>1</sub> (H <sub>2</sub> O) <sub>2.1667</sub>                                                                                             | 2.17                 |
|                  | CSHQ-TobD                                           | ((CaO) <sub>1.25</sub> (SiO <sub>2</sub> ) <sub>1</sub> (H <sub>2</sub> O) <sub>2.75</sub> ) <sub>0.6667</sub>                                                                            | 1.83                 |
|                  | CSHQ-TobH                                           | (CaO) <sub>0.6667</sub> (SiO <sub>2</sub> ) <sub>1</sub> (H <sub>2</sub> O) <sub>1.5</sub>                                                                                                | 1.5                  |
|                  | KSiOH                                               | ((KOH) <sub>2.5</sub> (SiO <sub>2</sub> ) <sub>1</sub> (H <sub>2</sub> O) <sub>0.2</sub>                                                                                                  | 0.2                  |
|                  | NaSiOH                                              | ((NaOH) <sub>2.5</sub> SiO <sub>2</sub> H <sub>2</sub> O) <sub>0.2</sub>                                                                                                                  | 0.2                  |

**Table 4**

Weighted average molar combined water content in C-S-H reported in the literature.

| Description                                                            | H <sub>2</sub> O content<br>(mol H <sub>2</sub> O per mol<br>C-S-H) | Source |
|------------------------------------------------------------------------|---------------------------------------------------------------------|--------|
| Freeze-dried C-S-H                                                     | 1.4–1.5                                                             | [33]   |
| “D-dried” (freeze dried) C-S-H                                         | 1.3                                                                 | [36]   |
| Saturated C-S-H globule                                                | 1.8                                                                 |        |
| Saturated C-S-H globule + monolayer water                              | 2.1                                                                 |        |
| Model fit #1 C-S-H                                                     | 3.14                                                                |        |
| Model fit #2 C-S-H                                                     | 3.53                                                                |        |
| C-S-H excluding gel water, between 7 and 300 days                      | 1.8                                                                 | [37]   |
| Stoichiometric (all combined water but no<br>adsorbed or liquid water) | 1.8                                                                 |        |
| D-dried                                                                | 1.4                                                                 | [38]   |
| CSH at 11% RH                                                          | 2.1                                                                 |        |
| Saturated gel, including liquid water between<br>particles             | 4                                                                   |        |

$$\sigma = 153.02 \bullet cwf - 22.59 \quad (4)$$

Eq. (4) is based on experimental data for limestone and limestone calcined clay mortars with clinker factor between 29 and 94 wt%, which has an  $R^2 > 0.96$  and 95% confidence interval of  $\pm 6.6$  MPa [43]. We compared the compressive strength performance results (Eq. (4)) to publicly available 28-day compressive strength test results per EN 196-1 of CEM I mortar samples (which uses a 1:3 cement to sand ratio by mass) from UK producers (Welsh Dragon, Holcim, Heidelberg, available S2 Table 3). The alite content in clinker for the UK producer CEM I mortar samples were included where reported from chemical analysis (e.g., XRD), and otherwise assumed to be in the range 50–75 wt% alite (Table 1).

## 2.6. Clinker carbon footprint

For each simulated clinker (per Section 2.2), we calculated CO<sub>2</sub>-eq. emissions from limestone calcination and fuel combustion to assess the effect of alite content on the carbon footprint of CEM I cement, since these are the two highest contributing sources (>90% of total CO<sub>2</sub> emissions for CEM I with 95 wt% clinker and 5 wt% gypsum [7]). We

**Table 5**

Phase stoichiometry, molar mass, and standard heat of formations for clinker phases and feedstock oxides. Values from [18] unless marked with \*, where values are from NIST [45].

| Phase             | Formula                                                                                     | Molar mass (g mol <sup>-1</sup> ) | ΔH <sub>f</sub> <sup>0</sup> (kJ mol <sup>-1</sup> ) |
|-------------------|---------------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------------------|
| Alite             | C <sub>3</sub> S (3CaO·SiO <sub>2</sub> )                                                   | 228                               | −2931                                                |
| Belite            | C <sub>2</sub> S (2CaO·SiO <sub>2</sub> )                                                   | 172                               | −2308                                                |
| Aluminate         | C <sub>3</sub> A (3CaO·Al <sub>2</sub> O <sub>3</sub> )                                     | 270                               | −3561                                                |
| Ferrite           | C <sub>4</sub> AF<br>(4CaO·Al <sub>2</sub> O <sub>3</sub> ·Fe <sub>2</sub> O <sub>3</sub> ) | 486                               | −5080                                                |
| Calcite           | CaCO <sub>3</sub>                                                                           | 100                               | −1207                                                |
| SiO <sub>2</sub>  | SiO <sub>2</sub>                                                                            | 60                                | −910.7                                               |
| Alumina*          | Al <sub>2</sub> O <sub>3</sub>                                                              | 102                               | −1675.7                                              |
| Iron oxide        | Fe <sub>2</sub> O <sub>3</sub>                                                              | 160                               | −862.2                                               |
| CaO*              | CaO                                                                                         | 56                                | −635.5                                               |
| CO <sub>2</sub> * | CO <sub>2</sub>                                                                             | 44                                | −393.5                                               |

assumed 100% of CaO was from limestone (modelled as CaCO<sub>3</sub> here). The IPCC method for clinker process emissions [44] was used to give limestone calcination emissions in kg CO<sub>2</sub>-eq. tonne<sup>-1</sup> clinker per Eq. (5).

$$\text{calcination emissions} = 0.785 \bullet 1000 \bullet \sum_i m_i \bullet n_{\text{CaO}}^{(i)} \bullet \frac{56.0774}{MM_i} \quad (5)$$

where, for each clinker phase  $i$ ,  $m_i$  is its mass fraction in clinker (–),  $n_{\text{CaO}}^{(i)}$  is its stoichiometric content of CaO (mol fraction), and  $MM_i$  is its molar mass (g mol<sup>-1</sup>) (Table 5).

Theoretical heats of reaction for each clinker phase (Table 6) were calculated from the standard heats of formation (Table 5). The heat of reaction for calcination of CaCO<sub>3</sub> to produce CaO for each phase (i.e., the calcination energy) was then added to determine the total heat of reaction for each phase, and these were used with the phase composition of each clinker to determine the enthalpy of clinker formation. A kiln efficiency of 50% was assumed to scale up the clinker enthalpy to total kiln energy consumption and account for sensible heating and heat losses. This efficiency was based on the reported theoretical minimum energy of clinker formation at a standard composition of 1.7–1.8 kJ g<sup>-1</sup> and the reported best available technology clinker kiln energy consumption of 3.0–3.8 kJ g<sup>-1</sup> [46]. Two additional fuel efficiency cases

**Table 6**

Standard heat of reaction (from  $\Delta H_r^0$  in Table 5) required to form each clinker phase. Calcination energy ( $\text{kJ mol}^{-1}$ ) is energy to calcine limestone per mol reaction product. Total heat of reaction for each phase is the sum of  $\Delta H_r^0$  for clinker phase formation and calcination), given in both  $\text{kJ mol}^{-1}$  and  $\text{kJ g}^{-1}$ .

| Overall reaction                                                                                                                                                | Clinker phase formation                                                                                                                       | Calcination                                             | $\Delta H_{r,\text{clinker phase formation}}^0$<br>( $\text{kJ mol}^{-1}$ ) | $\Delta H_{r,\text{calcination}}^0$<br>( $\text{kJ mol}^{-1}$ ) | $\Delta H_{r,\text{tot}}^0$<br>( $\text{kJ mol}^{-1}$ ) | $\Delta H_{r,\text{tot}}^0$ ( $\text{kJ g}^{-1}$ ) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------|
| $3\text{CaCO}_3 + \text{SiO}_2 \rightarrow 3\text{CaO} \cdot \text{SiO}_2 + 3\text{CO}_2$                                                                       | $3\text{CaO} + \text{SiO}_2 \rightarrow 3\text{CaO} \cdot \text{SiO}_2$                                                                       | $3\text{CaCO}_3 \rightarrow 3\text{CaO} + 3\text{CO}_2$ | -113.8                                                                      | 534                                                             | 420.2                                                   | 1.8                                                |
| $2\text{CaCO}_3 + \text{SiO}_2 \rightarrow 2\text{CaO} \cdot \text{SiO}_2 + 2\text{CO}_2$                                                                       | $2\text{CaO} + \text{SiO}_2 \rightarrow 2\text{CaO} \cdot \text{SiO}_2$                                                                       | $2\text{CaCO}_3 \rightarrow 2\text{CaO} + 2\text{CO}_2$ | -126.3                                                                      | 356                                                             | 229.7                                                   | 1.3                                                |
| $3\text{CaCO}_3 + \text{Al}_2\text{O}_3 \rightarrow 3\text{CaO} \cdot \text{Al}_2\text{O}_3 + 3\text{CO}_2$                                                     | $3\text{CaO} + \text{Al}_2\text{O}_3 \rightarrow 3\text{CaO} \cdot \text{Al}_2\text{O}_3$                                                     | $3\text{CaCO}_3 \rightarrow 3\text{CaO} + 3\text{CO}_2$ | 21.2                                                                        | 534                                                             | 555.2                                                   | 2.1                                                |
| $4\text{CaCO}_3 + \text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3 \rightarrow 4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{Fe}_2\text{O}_3 + 4\text{CO}_2$ | $4\text{CaO} + \text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3 \rightarrow 4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{Fe}_2\text{O}_3$ | $4\text{CaCO}_3 \rightarrow 4\text{CaO} + 4\text{CO}_2$ | -0.1                                                                        | 712                                                             | 711.9                                                   | 1.5                                                |

(low: 40% efficient, and high: 60% efficient) were considered as a sensitivity analysis, and included in Supporting Information S1, S6.0.

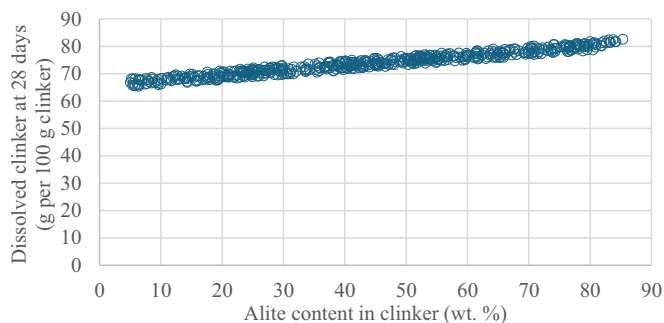
Fuel combustion emissions were calculated from the total kiln energy consumption for each clinker composition and the 2021 Global Cement and Concrete Association (GCCA) UK average carbon intensity of kiln fuel: 72.6 g  $\text{CO}_2$ -eq. emissions  $\text{MJ}^{-1}$  [47]. We additionally performed sensitivity analysis considering an upper-case fuel emission scenario of 108.8 g  $\text{CO}_2$ -eq. emissions  $\text{MJ}^{-1}$ ,  $1.5 \times$  the 2021 GGCA UK average carbon intensity. This represents a likely upper limit of fuel intensity, close to the highest fuel intensity reported in the GCCA report (104.2 g  $\text{CO}_2$ -eq. emissions  $\text{MJ}^{-1}$  per Austrian production in 1990), and is further discussed in Supporting Information S1, S6.0. We compared the calculated emissions to total  $\text{CO}_2$ -eq. emissions results (Environmental Product Declaration (EPD), A1–A3 category) for high strength (>52.5 N at 28-days curing) CEM I cements from the UK producers listed in Section 2.5 (refer to Supporting Information S2, Table 3).

### 3. Results and discussion

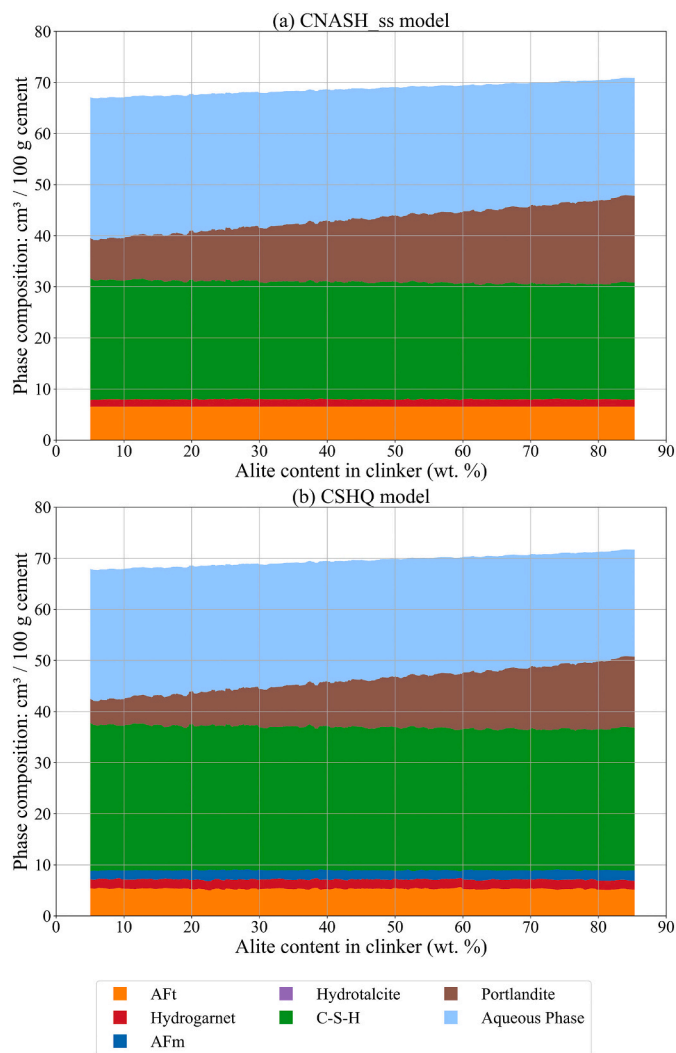
#### 3.1. Clinker dissolution and hydrated cement phase assemblage

Our results show that the total amount of dissolved clinker, proportional to the degree of hydration, increases with the alite content in clinker (Fig. 2). At 28-days of curing, approximately 82 wt% of the total clinker mass had dissolved for clinkers with high alite content (85 wt%), compared to 67 wt% dissolution for the same mass of low alite clinker (5 wt%). This is approximately a 5 wt% difference over the range of typical alite content in clinker (50–75 wt% alite in clinker, Table 1). The predicted amount of dissolved clinker at 28-days curing is conservative for the purpose of strength comparison (e.g., maximum reactivity predicted), particularly for high-belite clinkers (refer to Supporting Information SI, S1.0). While degree of hydration increases with alite content, the increase in mass is predominately from additional dissolved calcium in the alite, resulting in an increase in the bulk Ca/Si ratio.

Fig. 3 depicts the hydrated cement phase assemblage volumes per 100 g of CEM I cement, predicted using the thermodynamic model. Both



**Fig. 2.** Total dissolved clinker at 28-days curing for each clinker, compared with alite content in clinker (wt%).



**Fig. 3.** Volume ( $\text{cm}^3$ ) of hydrate phases formed in (a) CNASH<sub>ss</sub> model and (b) CSHQ model from 100 g cement hydrated with 50 g water at 28-days curing. Raw data has been smoothed for visualisation purposes using Savitzky-Golay smoothing via scipy package solvay\_filter with a window length of 15.

the CNASH<sub>ss</sub> and CSHQ models predicted C-S-H and portlandite as the dominant hydrate phases, with other hydrates (e.g., AFm, AFt, hydrogarnet) present in minor amounts. <0.03 mol of AFt phase, primarily as ettringite, forms in both models, with <0.003 mol of AFm phase primarily as monosulphate predicted by the CSHQ model. A small amount (<0.014 mol) of Fe-siliceous hydrogarnet is predicted due to the presence of Fe in Portland clinker. It is among the most stable solid phases for

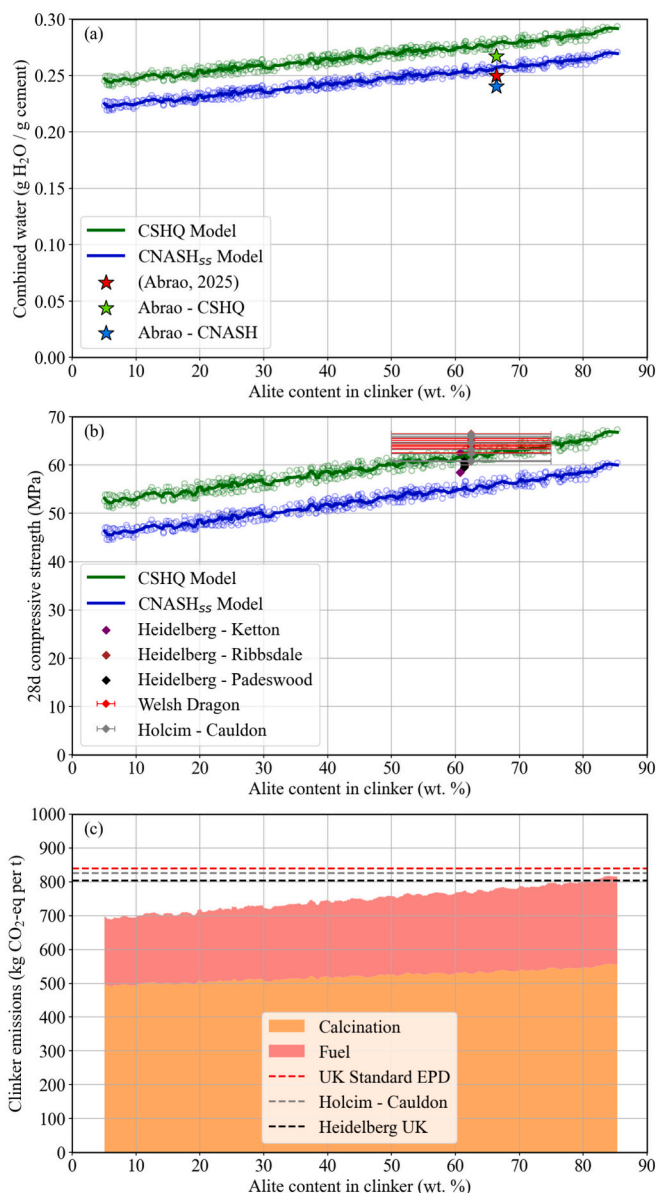
Fe in Portland cement systems. It has been reported to precipitate after 1 day at 20 °C in CEM I paste [48]. Detailed results for phase formation (e.g., molar amounts of each end member formed), are available in Supporting Information S2.

The predicted C-S-H in both models is approximately constant over the range of alite content, due to the increase in bulk Ca/Si ratio and preferential equilibrium formation of portlandite with increasing calcium availability. As expected (see Section 2.4), more C-S-H and less portlandite was predicted using the CSHQ model than the CNASH<sub>ss</sub> model. Correspondingly, the CNASH<sub>ss</sub> model predicted 9–11% more aqueous solution on a molar basis than the CSHQ model across the 5–85 wt% alite content in clinker. Further details describing the predicted C-S-H composition are available in Supporting Information S1, S3.0. Portlandite formation increased with alite content, 117% and 147% on a molar basis across the 5–85 wt% alite content in clinker range for the CNASH<sub>ss</sub> and CSHQ models respectively, which agrees with the simplified hydration reaction stoichiometry of alite and belite [49]. Portlandite has a low combined water content (1 mol H<sub>2</sub>O per mol portlandite, per Section 2.5 Table 3), and therefore has a lower contribution to cement strength performance per mol from the combined water fraction prediction, when compared to the other solid phases in the hydrated cement phase assemblage (e.g., C-S-H). This is in agreement with previously reported work [50], which investigated the relationship between compressive strength, hydration products, and gel-space ratio (e.g., the ratio of hydration product volume and the sum of hydration product volume and capillary porosity), and observed strength decreased with increasing portlandite in the phase assemblage, concluding that portlandite has a reduced ‘mechanical efficiency’ compared with other hydrates (e.g., C-S-H).

### 3.2. Combined water, cement strength performance, and carbon footprint

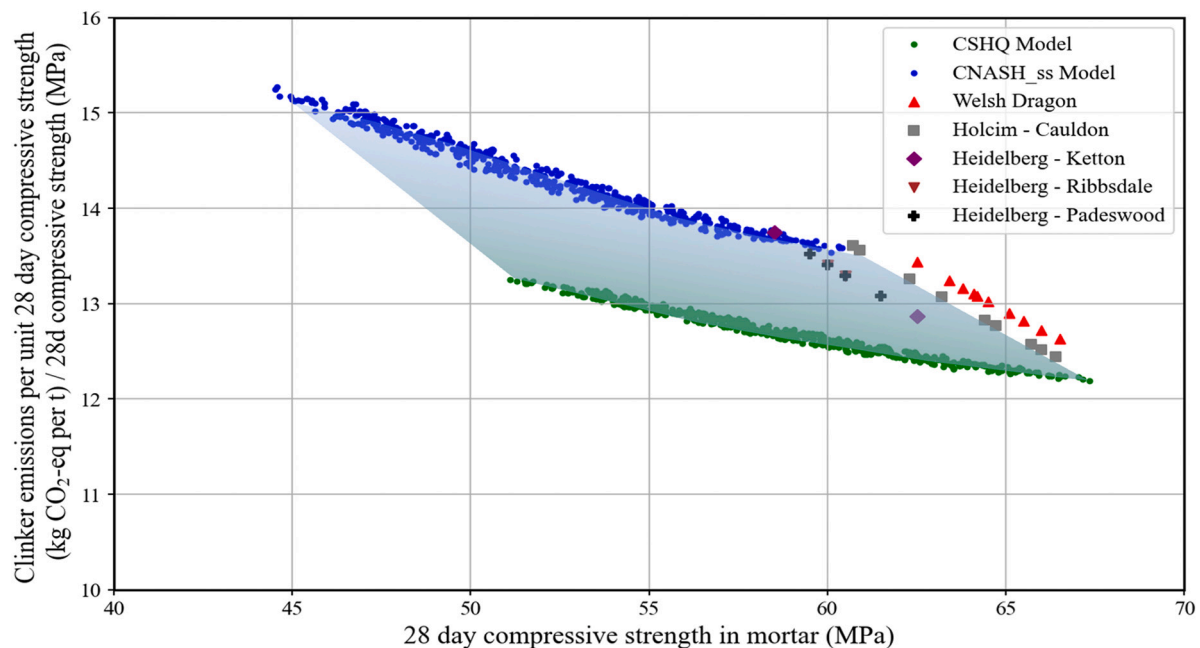
The hydrated phase assemblage at 28-days curing was used to calculate the combined water and the predict cement strength performance of the CEM I mortars. Fig. 4a shows the total combined water result contained in the hydrate phases at 28-days curing for both models, compared to the experimental result of Abrao et al. [22]. Overall, the modelled combined water increased from  $0.23 \pm 0.1$  to  $0.28 \pm 0.1$  g H<sub>2</sub>O per g cement with increased alite content in clinker, predominantly due to increased portlandite formation. The combined water for the 66 wt% alite clinker in CEM I as studied by Abrao et al. (e.g., [22]) is shown as a red star (experimental result), green star (CSHQ model result), and blue star (CNASH<sub>ss</sub> model result). The experimental combined water result (0.25 g H<sub>2</sub>O per g cement) from [22] falls between the two model predicted results (0.24 and 0.267 g H<sub>2</sub>O per g cement), with an error below 7%. Thus, the variance in combined water (18.9%–20.4%) across the range of alite content in clinker was larger than the error in the model (<7%), and therefore alite content in clinker is significant to the combined water content in the mortar studied.

The predicted cement strength performance for the modelled mortars is shown in Fig. 4b. Cement strength performance increased with clinker alite content from ~52–67 MPa for the CSHQ model, and ~46–61 MPa for the CNASH<sub>ss</sub> model, which is a 1.5 MPa strength increase of the mortar for each additional 10 wt% alite in clinker, for mortars with 0.5 w/c ratio. The strength results from UK site based CEM I mortar were closer aligned with the results of the CSHQ model. This is expected as the CSHQ model is more accurate than CNASH<sub>ss</sub> for the high Ca/Si molar ratios typical of CEM I cements [25]. Fig. 4c shows the calcination and fuel combustion CO<sub>2</sub>-eq. emissions contributions to clinker carbon footprint, where the highest alite content clinkers (85 wt % alite, up to 830 kg CO<sub>2</sub>-eq. emission per kg clinker) had 18.6% higher CO<sub>2</sub>-eq. emissions than the lowest alite content clinkers (5 wt% alite, down to 700 kg CO<sub>2</sub>-eq. emission per kg clinker). With increasing alite content across the 5–85 wt% alite in clinker range, calcination related emissions increased from 500 to 550 kg CO<sub>2</sub>-eq. per tonne clinker, and fuel related emissions increased from 200 to 280 kg CO<sub>2</sub>-eq. per tonne



**Fig. 4.** (a) Combined water of CEM I mortar at 28-days curing with clinker of varying alite content, (b) predicted compressive strength of CEM I mortar at 28-days curing and (c) calculated calcination (orange band) and fuel (pink band) related CO<sub>2</sub>-eq. emissions per tonne clinker against alite content in clinker (wt %). (a) and (b) both show raw model results from CNASH<sub>ss</sub> (blue unfilled dots) and CSHQ (green unfilled dots) models for C-S-H, with a smoothed line added for visibility improvement using Savitzky-Golay smoothing via scipy package solvay\_filter with a window length of 15. Equivalent Savitzky-Golay smoothing has also been applied to (c). Result of (a) is compared to the experimental result from Abrao, 2025 (red star) [22], CSHQ modelled result (green star) and CNASH<sub>ss</sub> modelled result (blue star). Results in (b) compared to 28-day strength results for CEM I mortars from UK producers (Welsh Dragon, Heidelberg and Holcim). Where alite content in clinker was not reported, error bars extend horizontally over the typical 50–75 wt% alite content in clinker. The relevant UK EPD A1-A3 total global warming potential results for CEM I 52.5 N cements are shown on (c) (UK standard, Holcim EPD for Cauldon site, and Heidelberg UK). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

clinker. Based on the sensitivity analysis (Supporting Information S1, S6.0), the worst-case scenario of a low efficiency kiln (40%) and  $1.5 \times$  UK GCCA average emissions, resulted in a maximum 23.8% change over the range of alite content in total clinker emissions. This is less than the



**Fig. 5.** Clinker emissions per unit 28-day compressive strength in CEM I mortar of clinkers with varying alite content against their 28-day cement strength performance in mortar. The calculated emissions (in kg CO<sub>2</sub>-eq. t<sup>-1</sup> clinker) from Fig. 4b were divided by the predicted cement strength performance (in MPa at 28-days curing) from Fig. 4a for each mortar. The shaded section indicates the region predicted by the CSHQ (green) and CNASH<sub>ss</sub> (blue) models. Site data (Welsh Dragon, Holcim, Heidelberg) is normalised using the most relevant EPD (UK standard for Welsh Dragon, Holcim Cauldon for Cauldon, and Heidelberg UK standard for Ketton, Ribbsdale, and Padeswood). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

28.8% change in strength over the range of alite content based on CSHQ model case. The CO<sub>2</sub> released from calcination had the highest contribution to the clinker emissions. The calculated clinker emissions represent >90% of the footprint reported by the UK CEM I EPDs at typical alite in clinker contents (~60 wt%), which justifies the simplified environmental assessment approach taken.

Both cement strength performance and carbon footprint increased with alite content, demonstrating the need for a performance-based functional unit to determine the optimal clinker composition. Fig. 5 below shows the performance-based emissions (kg CO<sub>2</sub>-eq. emissions t<sup>-1</sup> clinker MPa<sup>-1</sup> strength at 28-days curing) of CEM I mortar made with clinker of varying alite content.

For CEM I mortars at 0.5 w/c ratio, Fig. 5 shows that the minimum carbon footprint per strength occurs at the highest cement strength performance, which correspond to the clinkers with the highest alite content per Fig. 4b. Overall, this corresponds to an 11–13% decrease (1.5–2 kg CO<sub>2</sub>-eq. tonne<sup>-1</sup> MPa<sup>-1</sup> strength at 28-days curing) in strength performance-based emissions over the 15 MPa range of predicted strength. The data from UK cement producers fall within the predicted range bounded by the CNASH<sub>ss</sub> and CSHQ model results, varying up to ±0.6 kg CO<sub>2</sub>-eq. tonne<sup>-1</sup> MPa<sup>-1</sup> strength at 28-days curing over ±3 MPa strength at 28-days curing. This variance is primarily due to differences in the measured cement strength performance of individual samples. The variance in the site data is higher than the model, as the EPD-based emissions data (kg CO<sub>2</sub>-eq. per tonne) is fixed per site and does not vary with composition. Overall, our result agrees with Scrivener et al. (2023) that low-alite clinkers are environmentally unfavourable, as the reduction in emissions does not compensate for the loss in performance [35], and demonstrates the importance of both clinker composition and performance-based LCA for cement and concrete.

#### 4. Insights for low carbon cement production

##### 4.1. Clinker substitution

Our results clearly show that clinker composition is important for

optimising the cement strength performance in low-carbon cements. While high alite clinkers had the lowest carbon footprint on a cement strength performance basis, the use of high clinker content CEM I cement is increasingly uncommon in concrete. Increasing the use of low clinker composite cements is a key decarbonisation strategy for the cement industry (e.g., GCCA [8]), and the most realistic strategy for deep decarbonisation [51]. In composite cements, portlandite is consumed through reaction with pozzolanic materials (e.g., siliceous clinker substitutes like calcined clay) to produce C-S-H. High alite content clinker can enable production of composite cement with low clinker factor without compromising strength performance both due to the availability of portlandite for the pozzolanic reaction, and due to the higher dissolution rate of alite (Fig. 2). Higher alite clinkers are therefore likely to improve the cement strength performance of low-clinker composite cements used in concrete. This result should be tested in future experimental studies of concretes.

The combined water model proposed in this work correlates the strength of mortar to the type and amount of hydration products formed in the phase assemblage, rather than the type of cement used. The empirically derived combined water fraction to cement strength performance correlation (Eq.(4)) included data from composite cements cured for 7–91 days with up to 70% clinker substitution, including calcined clay cements with a high alumina content, with 95% confidence interval of ±6.6 MPa [22] compared to the ±5 MPa variation observed in a standardised interlaboratory test for a single cement [42]. We therefore expect that the combined water fraction model is suitable for use for the purpose for life cycle assessment of composite cements with C-S-H (or C-(A-)S-H) as the primary hydration product. Future studies should expand the cement strength performance-based environmental assessment presented here to composite cements with various clinker substitutes (e.g., GBFS, limestone calcined clay (“LC3”)) at high clinker substitution rates (e.g., <50 wt% clinker in cement), to identify low clinker composite cements with adequate cement strength performance enabled by producing high-alite clinker.

#### 4.2. Kiln fuel ash

Incorporating clinker composition and performance-based functional units into LCA studies of cement has further implications on the decarbonisation strategy for the cement industry. For example, current cement industry decarbonisation strategies [8,52–55] prioritise substitution of fossil fuels in clinker production (e.g., with lower carbon footprint waste fuels, which can also introduce minor species and affect clinker phase formation, discussed in Section 4.3). However, fuel type affects clinker phase composition, as the ashes produced from combustion are combined into the clinker [3,56]. The required ratios of CaO, SiO<sub>2</sub>, Al<sub>2</sub>O<sub>3</sub> and Fe<sub>2</sub>O<sub>3</sub> to achieve target phase composition in the clinker are given by the kiln feedstock proportioning targets (Eqs. (6–8), Table 7).

Solid fuels for clinker production (e.g., coal) typically produce a siliceous ash [56,58], which decreases the lime saturation factor (Eq. (6)) and increases the silica ratio (Eq. (7)). This promotes belite formation over alite, which we have demonstrated to be undesirable for cement strength performance and may hinder clinker performance in low clinker composite cement. The kiln feed can be corrected with additional Ca source (e.g., limestone) to account for fuel ash and maintain the lime saturation factor in the required range [56], however this additional Ca will increase the calcination emissions per tonne of clinker for the same alite content. Therefore, the relationship between fuel ash and kiln feed Ca content, clinker composition, and cement strength performance, should be included in the evaluation of lower carbon footprint cement strategies involving alternative fuels and low clinker composite cement. This can be done using a thermochemical model of the kiln chemistry to predict alite content in clinker (e.g., [59]). An example use case is quantifying the reduction in calcination emissions and increase in cement strength performance of reducing siliceous fuel ash in clinker production (e.g., by using fuels with low ash formation such as natural gas). In any case, our results call into question the benefits of some alternative fuel use in clinker production (especially those that produce higher ash content, and more siliceous ash, when burned), and that the decarbonisation strategy for the cement industry should be revised taking this insight into account.

#### 4.3. Ca sources and influence of diverse input chemistries on clinker quality

In this paper we have thus far assumed 100% of the Ca in the clinker feed is derived from limestone, which is the highest source of CO<sub>2</sub>-eq. emissions in CEM I production. The use of alternative kiln feedstocks as raw materials for clinker production is a lesser-implemented decarbonisation strategy for cement. Of particular interest are decarbonated sources of Ca (e.g., CaO rich slags), since they supply Ca without the CO<sub>2</sub> emissions associated with limestone calcination. Available decarbonated sources of Ca can displace 20–30% of limestone in kiln feed [13,60], potentially avoiding 100–165 kg CO<sub>2</sub> of calcination emissions per tonne of clinker. The advantage of reducing calcination emissions using decarbonated sources of Ca is that the subsequent cement carbon

footprint can be further reduced by clinker substitution. However, the availability of these materials is limited compared to the global demand for Portland cement products, and some are already fully used as clinker substitutes. Identifying the most effective use case of decarbonated sources of Ca is therefore imperative considering their constrained supply [13]. This is another use case for the expanded thermochemical model suggested in Section 4.2, since it could quantify which application (e.g., as kiln feed or clinker substitute) of these valuable materials minimises the overall carbon footprint per unit of cement strength performance gained in composite cements.

However, assessing alternative materials and alternative fuels with diverse chemistries is constrained by the limited availability of high-temperature thermodynamic data for clinkerisation reactions. For example, both alternative fuels and materials considered for clinker production are often wastes or industrial by-products, and can contain minor components which inhibit cement hydration (e.g., zinc [61]), or inhibit alite formation by stabilising belite (e.g., phosphorous [62]). Existing data do not comprehensively account for the influence of minor oxides, heavy metals, and other contaminants (e.g., Mg, Ti, F, Cl, Zn, P, Na, K, Sr, and Br [59]) on clinker formation [2,4]. Improved thermodynamic data, or machine learning based material discovery approaches, are essential for determining the maximum amounts of contaminated material which can be incorporated in clinker, and performance-based life cycle assessments.

#### 4.4. Practical limit for alite content in clinker

Our synthetic dataset contained clinkers with composition between 5 and 95 wt% each of alite and belite, 4–5 wt% each of aluminate and ferrite, and a maximum 5 wt% of inert phase. This resulted in samples up to ~85 wt% alite content, which exceeds the ~75 wt% practical limit of alite formation in a rotary kiln. This limit relates to the melt phase formed during clinker burning (which solidifies as the interstitial phase containing aluminate and ferrite). The melt phase is required to facilitate mass transfer and produce clinker nodules, which are needed to allow air-cooling of clinker after exiting the kiln. The quality of raw materials also contributes to the amount of interstitial phase formed (e.g., high iron clays, impure limestone deposits), with lower quality materials resulting in lower alite content in clinker due to impurities. However, high alite content clinkers (up to 100 wt% with high purity materials e.g., [63]) can be synthesised under laboratory conditions. We recommend prioritising development of novel process arrangements to increase clinker alite content beyond the limits of rotary kilns in future research.

In addition to the technical limitations to produce high alite clinker, it is also valuable to discuss the implications of the 28-day strength functional unit used in this analysis. The compressive strength at 28-days curing is the basis for cement specification for most applications [26,27] and previous strength-performance environmental assessments (e.g., [20]), as early-age strength is a particular concern for construction productivity (e.g., removal of formwork). However, high-belite cements typically have improved strength performance at later ages (e.g., [50,64,65]) likely due to the higher proportion of C-S-H formed from the hydration of belite, as well as improved performance against sulphate and chloride ingress due to lower portlandite content [64]. As a result, strength-weighted environmental analysis is sensitive to the curing time considered. However, the use of supplementary cementitious materials have demonstrated similar strength and durability enhancement at later ages [66], also likely due to additional C-S-H formation from portlandite as a result of the pozzolanic reaction. Therefore, the use of high-alite clinkers is recommended in the context of enabling lower clinker composite cements, where the use of supplementary cementitious materials could effectively replace the function of belite (e.g., for late age strength and durability enhancement) for lower emissions.

**Table 7**

Clinker feedstock proportioning targets and Bogue prediction for alite formation [56,57], using oxide CaO, SiO<sub>2</sub>, Al<sub>2</sub>O<sub>3</sub>, Fe<sub>2</sub>O<sub>3</sub> composition in wt%. Targets are dimensionless, and the target values given are examples based on conventional clinker composition (e.g., 50–75 wt% alite).

| Parameter                    | Target    | Definition (lte in wt%)                                                                                    |     |
|------------------------------|-----------|------------------------------------------------------------------------------------------------------------|-----|
| Lime saturation factor (LSF) | 0.95–0.97 | LSF = $\frac{\text{CaO\%}}{2.8\text{SiO}_2\% + 1.18\text{Al}_2\text{O}_3\% + 0.65\text{Fe}_2\text{O}_3\%}$ | (6) |
| Silica ratio (SR)            | 2.4–2.6   | SR = $\frac{\text{SiO}_2\%}{\text{Al}_2\text{O}_3\% + \text{Fe}_2\text{O}_3\%}$                            | (7) |
| Alumina ratio (AR)           | 1.5–1.8   | AR = $\frac{\text{Al}_2\text{O}_3\%}{\text{Fe}_2\text{O}_3\%}$                                             | (8) |

## 5. Conclusion

This work studied the effect of alite content in clinker on the strength and environmental performance of CEM I cement. Our results show that high-alite clinkers provide a dual benefit for producing low carbon cements for a given strength performance: high alite (85 wt% alite) clinkers are more reactive, with 15 wt% more clinker dissolved at 28 days than low alite (5 wt% alite) clinkers; and high alite clinkers produce more portlandite in the hydrated phase assemblage, which can support higher rates of clinker substitution with pozzolanic materials in composite cement. Alite contributes to superior cement strength performance (1.5 MPa of additional strength gain in CEM I mortar at 28 days per 10 wt% alite in clinker) than belite, and the highest alite clinkers had 28.8% higher cement strength performance (e.g., 67 MPa compared to 52 MPa) in CEM I mortar at 28 days than low alite clinker. Over the same range, the carbon footprint per mass when using high alite clinker was up to 18.6% higher (~830 kg CO<sub>2</sub>-eq. per tonne) compared to the lowest alite clinkers (down to ~700 kg CO<sub>2</sub>-eq. per tonne). However, when compared on a strength and mass basis, CEM I mortars made with the high alite clinkers had up to ~13% lower CO<sub>2</sub>-eq. emissions per tonne clinker per MPa of 28-day compressive strength, and further enable lower-clinker composite cements with adequate strength performance. We found that the combined water result calculated from thermodynamic binder chemistry has good agreement with experimental results, indicating that it is an efficient method to explore the performance of the cement and concrete design space. Performance-based LCA is essential to devise and accurately evaluate effective decarbonisation strategies for the cement and concrete industry.

## CRedit authorship contribution statement

**Pippa Edwards:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wilson Ricardo Leal da Silva:** Writing – review & editing, Supervision, Funding acquisition. **Paul Fennell:** Writing – review & editing, Supervision. **Rupert J. Myers:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare no competing financial interests.

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## Appendix A. Supplementary data

Supporting Information S1 contains model results (dissolution model, thermodynamic model), and further details of validating the combined water content against reported experimental results. The synthetic clinker database, cement input for GEMS, and site strength data is available in Supporting Information S2 (as a Microsoft Excel file). Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cemconres.2026.108217>.

## Data availability

Data will be made available on request.

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